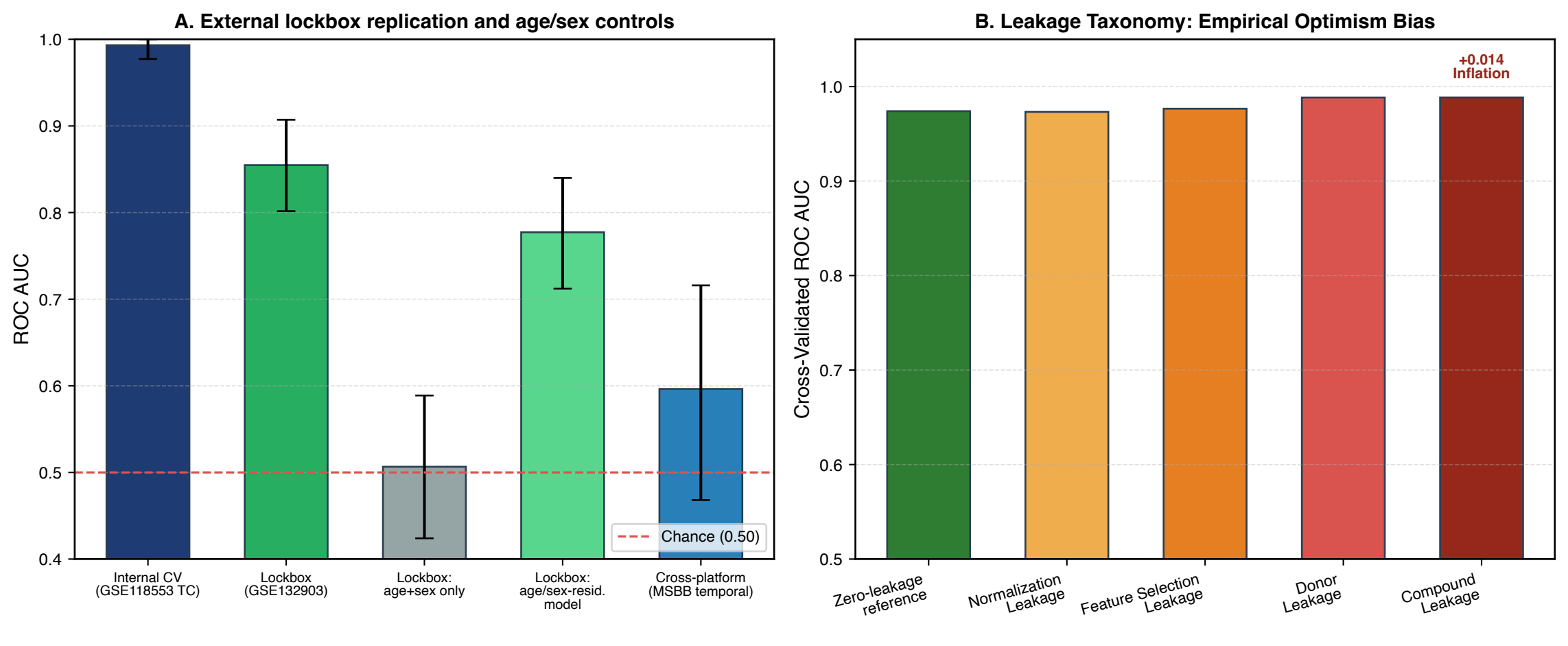
A. External lockbox replication and age/sex controls
ROC AUC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Chance (0.50)
Internal CV (GSE118553 TC)
Lockbox (GSE132903)
Lockbox: age+sex only
Lockbox: age/sex-resid. model
Cross-platform (MSBB temporal)
B. Leakage Taxonomy: Empirical Optimism Bias
Cross-Validated ROC AUC
+0.014 Inflation
1.0
0.9
0.8
0.7
0.6
0.5
Zero-leakage reference
Normalization Leakage
Feature Selection Leakage
Donor Leakage
Compound Leakage